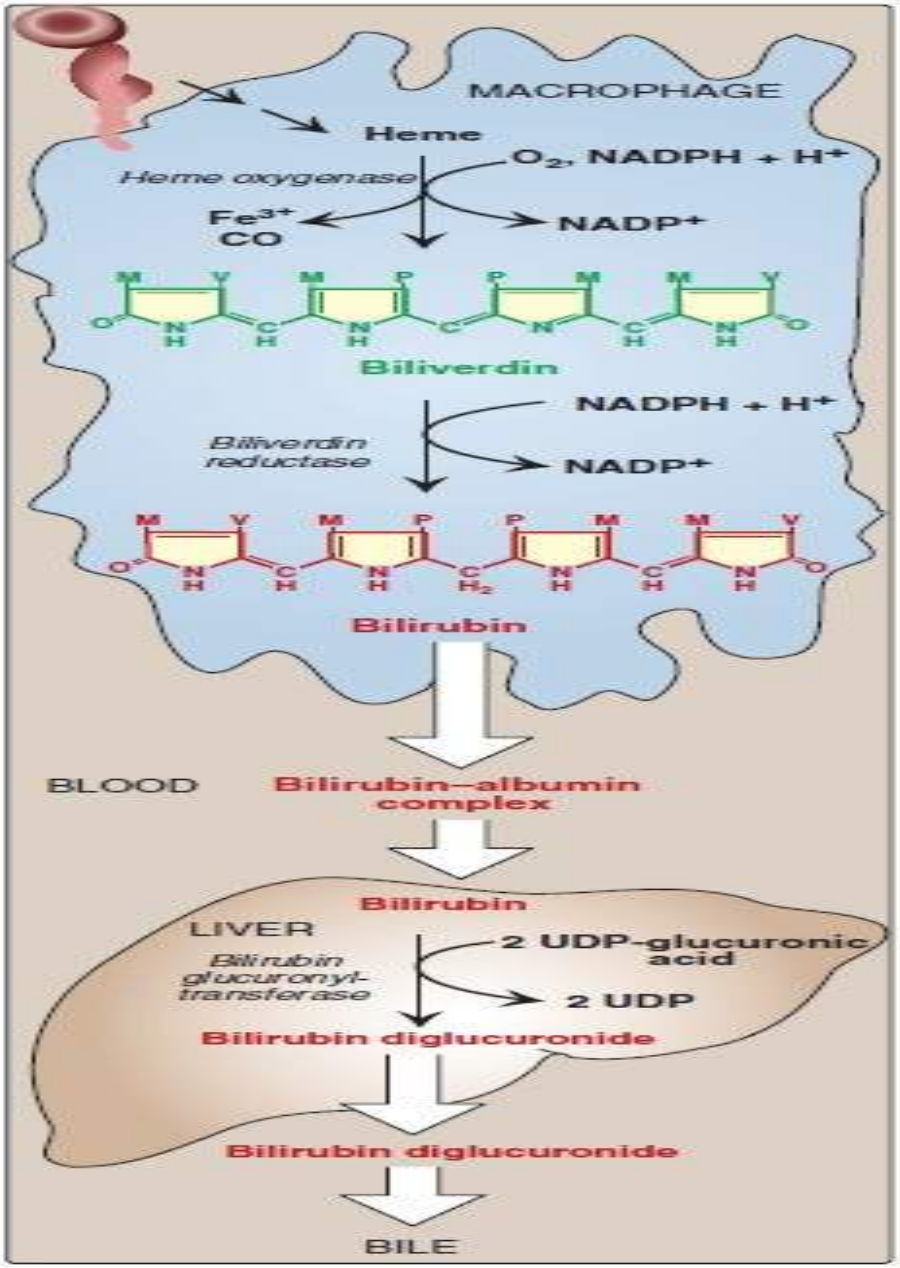

بسم الله الرحمن الرحيم

DEGRADATION OF HEME & BILIRUBIN METABOLISM

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Degradation of heme

After approximately 120 days in the circulation, red blood cells are taken up and degraded by the reticuloendothelial system, particularly in the liver and spleen (Figure 21.9). Approximately 85% of heme destined for degradation comes from senescent red blood cells, and 15% is from turnover of immature red blood cells and cytochromes from nonerythroid tissues.


MACROPHAGE
Heme
O_2, NADPH + H^+
Heme oxygenase
Fe^{3+}
CO
$NADP^+$
Biliverdin
NADPH + H^+
Biliverdin reductase
$NADP^+$
Bilirubin
BLOOD
Bilirubin–albumin complex
Bilirubin
LIVER
2 UDP-glucuronic acid
Bilirubin glucuronyl-transferase
2 UDP
Bilirubin diglucuronide
Bilirubin diglucuronide
BILE

Biosynthesis of heme

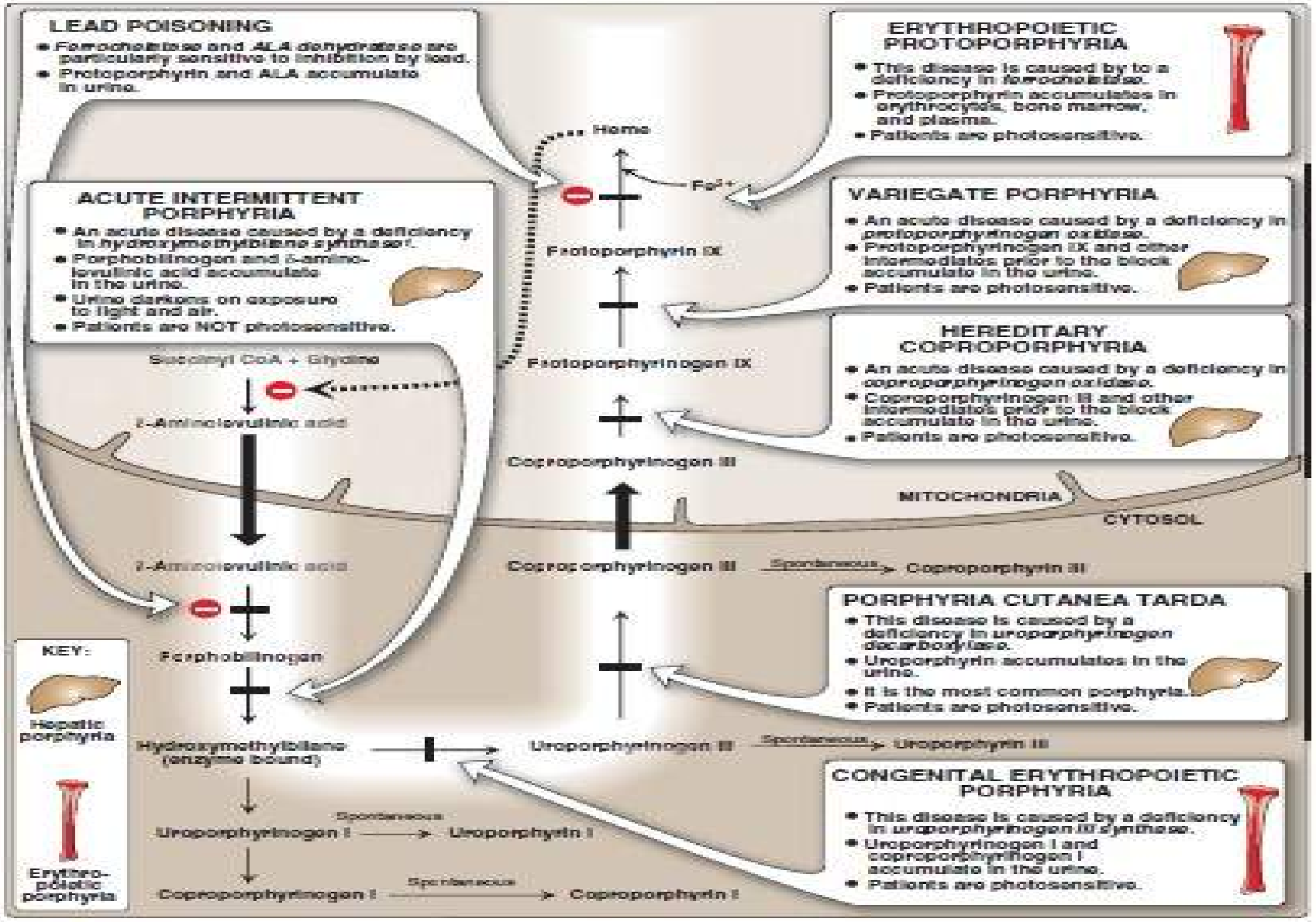


Figure 21.8
Summary of heme synthesis. [1]Also referred to as *porphobilinogen deaminase*.

The major sites of heme biosynthesis are the liver, which synthesizes a number of heme proteins (particularly cytochrome proteins), and the erythrocyte-producing cells of the bone marrow, which are active in hemoglobin synthesis. [Note: Over 85% of all heme synthesis occurs in erythroid tissue.]

In the liver, the rate of heme synthesis is highly variable, responding to alterations in the cellular heme pool caused by fluctuating demands for heme proteins. In contrast, heme synthesis in erythroid cells is relatively constant, and is matched to the rate of globin synthesis. The initial reaction and the last three steps in the formation of porphyrins occur in mitochondria,

whereas the intermediate steps of the biosynthetic pathway occur in the cytosol .Mature red blood cells lack mitochondria and are unable to synthesize heme.

Formation of δ-aminolevulinic acid (ALA):

All the carbon and nitrogen atoms of the porphyrin molecule are provided by glycine (a nonessential amino acid) and succinylcoenzyme A (an intermediate in the citric acid cycle) that condense to form ALA in a reaction catalyzed by ALA synthase (ALAS)

This reaction requires pyridoxal phosphate (PLP) as a coenzyme, and is the committed and rate-limiting step in porphyrin biosynthesis. [Note: There are two isoforms of ALAS, 1 and 2, each controlled by different mechanisms. Erythroid tissue produces only ALAS2, the gene for which is located on the X-chromosome. Loss of function mutations in ALAS2 result in X-linked sideroblastic anemia.]

End-product inhibition of ALAS1 by hemin

When porphyrin production exceeds the availability of the apoproteins that require it, heme accumulates and is converted to hemin by the oxidation of Fe2+ to Fe3+. Hemin decreases the activity of hepatic ALAS1 by causing decreased synthesis of the enzyme, through inhibition of mRNA synthesis and use (heme decreases stability of the mRNA), and by inhibiting mitochondrial import of the enzyme. [Note: In erythroid cells, ALAS2 is controlled by the availability of intracellular iron.

Effect of drugs on ALA synthase activity

Administration of any of a large number of drugs results in a significant increase in hepatic ALAS1 activity. These drugs are metabolized by the micro somal cyto chrome monooxygenase system—a heme protein oxidase system found in the liver . In response to these drugs, the synthesis of cytochrome

proteins increases, leading to an enhanced consumption of heme—a component of cytochrome proteins. This, in turn, causes a decrease in the concentration of heme in liver cells. The lower intracellular heme concentration leads to an increase in the synthesis of ALAS1 (derepression), and prompts a corresponding increase in ALA synthesis

Formation of porphobilinogen

The condensation of two molecules of ALA to form porphobilinogen by Zn-containing ALA dehydratase (porphobilinogen synthase) is extremely sensitive to inhibition by heavy metal ions, for example, lead that replace the zinc This inhibition is, in part, responsible for the elevation in ALA and the anemia seen in lead poisoning

Formation of uroporphyrinogen

The condensation of four porphobilinogens produces the linear tetrapyrrole, hydroxymethyl -bilane, which is isomerized and cyclized by uroporphyrinogen IIIsynthase to produce the asymmetric uroporphyrinogen III. This cyclic tetrapyrrole undergoes decarboxylation of its acetate groups, generating coproporphyrinogen III .These reactions occur in the cytosol.

Formation of heme

Coproporphyrinogen III enters the mitochondrion, and two propionate side chains are decarboxylated to vinyl groups generating protoporphyrinogen IX, which is oxidized to protoporphyrin IX. The introduction of iron (as Fe2+) into protoporphyrin IX occurs spontaneously, but the rate is enhanced by ferro - chelatase, an enzyme that, like ALA dehydratase, is inhibited by lead

Thank you

